



A TOPIC-CENTRED SENTIMENT ANALYSIS AND VISUALIZATION SYSTEM FOR MALAWIAN FACEBOOK CHATTER

SYSTEMS REQUIREMENTS SPECIFICATION

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Abstract

This project presents the design and implementation of a topic-centred sentiment analysis and visualization system tailored to Facebook conversations in Malawi. Recognizing that Facebook is the primary platform for public discourse in the Malawian digital ecosystem, the system focuses on transforming large volumes of unstructured posts and comments into meaningful, time-based insights about national issues. The project addresses key challenges in local text analytics, including English–Chichewa code-mixing, informal slang, emoji’s, and the scarcity of labeled datasets for low-resource languages. Public data are collected from selected Malawian Facebook pages through a tool [1] available on the online platform apify. A preprocessing pipeline cleans and normalizes multilingual text before applying supervised learning techniques for topic classification and sentiment analysis. Posts are first assigned to predefined national topics, after which both posts and their comments are classified into positive, neutral, or negative sentiment categories. The system combines traditional machine learning models with a fine-tuned transformer model based on AfriBERTa to improve accuracy in handling localized language patterns. Processed data are stored in a database and presented through an interactive dashboard developed using Streamlit or Dash, enabling visualization of topic-wise sentiment trends over time. Model performance is evaluated using accuracy, precision, recall, F1-score, and human validation through inter-annotator agreement. By converting Facebook chatter into interpretable visual analytics, the system supports policymakers, researchers, journalists, and civil society organizations in understanding public opinion on governance, social issues, and national events. The project contributes to African Natural Language Processing research by developing a localized, platform-specific, and language-aware sentiment analysis pipeline for Malawi.

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1. Introduction

1.1 Background

The rapid global expansion of social media has fundamentally changed how citizens communicate and share opinions on national events [2]. In Malawi, Facebook has emerged as the dominant platform for public dialogue [3] surpassing other networks like Twitter and Instagram in user engagement. It serves as a primary hub where Malawians actively discuss a wide range of topics, including politics, the economy, social issues, and daily life, making it a priceless source of public opinion data. Despite the vast amount of data, the enormous volume and unstructured nature of these conversations make manual interpretation practically impossible for [4,5]. Furthermore, the Malawian linguistic landscape on social media presents significant challenges for standard analytical tools. Conversations often involve a complex mix of English and Chichewa (code-switching), along with local slang (such as "thaffe" or "2bad") and abbreviations. Example posts are such as those made by times360, zodiac, malawi24 and other NGO'S pages. Existing automated sentiment analysis tools, which are primarily designed for Western contexts and standard , struggle to capture the nuances of localized expressions and informal text [6, 7]. As the result there is a struggle to capture the nuances of localized expressions and informal text, which limits the ability of researchers and policymakers to derive actionable insights from online discourse. Consequently, there is a critical need for a localized, automated system specifically modified for the Malawian context in chichewa. In order to transform unstructured social media text into clear, interpretable insights, as it aims to support evidence-based decision-making for policymakers, journalists, and non-governmental organisations (NGOs) across the [8, 9]. Furthermore, the system can serve as a valuable tool for improving communication strategies, evaluating public response to interventions, and supporting evidence-based planning and governance in Malawi.

1.2 Problem Statement

Facebook is the dominant platform for public dialogue in Malawi, yet it remains significantly understudied compared to other social media, and the vast volume and high velocity of its conversations make manual analysis practically impossible. Current automated sentiment analysis tools primarily designed for Western contexts and standard English, not suitable for the malawian linguistic ,which is characterized by complex English-Chichewa code-switching, informal slang (such as "thaffe" and "2bad") and culturally specific expressions. Furthermore, there is limited local research combining topic classification with sentiment analysis to generate time-based insights for decision-makers. This gap prevents policymakers, researchers, and NGOs from deriving actionable insights from online public opinion, thereby hindering evidence-based decision-making on national issues.

1.3 Scope of the Development Project

1.3.1 Features Included

Table 1: Included Features

Feature	Description
Sentiment Classification	Assigns Positive, Neutral, or Negative labels using ML and transformer models.
Dashboards with Visual Analytics	Includes time-series trends and example posts for context.
Data Pre-processing Pipeline	Handles emojis, URLs, code-mixing, and local slang normalization.
Model Training and Evaluation	Uses Logistic Regression, Naïve Bayes, BERT, RoBERTa, AfriBERTa with Accuracy, Precision, Recall, F1-score.
Data Collection Module	Collects publicly available Facebook posts and comments related to selected topics in Malawi.

1.3.2 Features Excluded

Table 2: Excluded features

Excluded Feature	Reason
Real-time Data Ingestion	Data will be collected periodically, not as a live stream.
Mobile Offline Mode	System is web-based and requires internet access.
Traditional Deep Learning Models (CNNs, LSTMs)	Require large datasets not available for Chichewa.
Multi-platform Social Media Integration	The project will focus only on Facebook data and will not include platforms such as TikTok, Instagram, or WhatsApp.

1.3.3 Operational Setting

1.3.3.1 Current System

Currently there is no record of an automated system specifically designed to analyze the Malawian Facebook landscape. Monitoring of public opinion is primarily conducted through manual

interpretation of posts and comments on pages such as Times360, Zodiak, and Malawi24. This manual process is deemed impractical and practically impossible due to the massive volume and high velocity of social media data, as well as the linguistic complexities of English-Chichewa code-switching and local slang.

1.3.3.2 System Integration

The system is designed to integrate into a modern data science workflow using the following tools and interfaces:

Table 3: System Integrations

Tool / Interface	Description
Apify	Used for web scraping Facebook posts and comments.
Data Export Formats (CSV / Excel / JSON)	Supports flexible data import/export.
Database Management Systems(PostgreSQL/MongoDB)	Stores structured text, timestamps, and sentiment labels.
Web Dashboards	Provides interactive dashboards for visualization.
Decision-Making Workflows	Supports policymakers and NGOs with actionable insights.

1.4 System Personnel

Table 4: System Personnel

Role	Description
System Users	NGOs, media practitioners, lecturers, and advisors using the dashboard.
Developers	Alfred Banda, Fitina Sichali, Viwongo Chima, Thando Mwandira, Rahema Likongaji, Perpetual Judith Chawanje.
System Owner	Department of ICT, Mzuzu University.

2. System Requirements

2.1 System Overview

System Purpose

The system is a sentiment-classification pipeline developed for Facebook comments to address the absence of localized natural language processing tools for Malawian social media discourse.

Public Facebook pages in Malawi generate large volumes of multilingual comments written in Chichewa, English, and mixed variants that existing sentiment analysis tools (designed primarily for Western and high-resource language contexts) are unable to process reliably. The system provides an automated, reproducible pipeline for classifying such comments into three sentiment categories: *positive*, *neutral*, and *negative*. In doing so, it supports policymakers, researchers, media practitioners, and civil society organizations in deriving evidence-based insights from online public opinion.

System Components

The system is composed of the following principal components:

- **Data Ingestion Module** - Retrieves Facebook posts and comments.
- **Preprocessing Module** - Cleans and normalizes ingested text prior to classification
- **Classification Module** - Applies pre-trained sentiment models to preprocessed records
- **Database Backend** - The central data store for all raw data, processed outputs, and meta-data.
- **Visualization Dashboard** - Provides an interactive interface for exploring sentiment trends and topic distributions.

System Workflow

The system operates as a database-centered pipeline. Public Facebook posts and comments are sourced from approved pages and stored in the database. The preprocessing module retrieves raw records, cleans and normalizes the text, and writes the processed data back to the database. The classification module applies trained models to assign topic labels to posts and sentiment labels to comments. All results are uploaded to the database. The visualization dashboard retrieves processed data to present topic-wise sentiment trends, distributions, and example content to end users. The model evaluation runs periodically, and performance metrics are stored for monitoring purposes.

2.2 Functional Requirements

The table below specifies the functional requirements of the delivered system. Each requirement describes what the system shall do for its users and operators, independent of implementation detail.

Table 5: Functional Requirements

Req. No	Requirement	Description
FR1	Collect public Facebook posts and comments automatically.	The system shall retrieve posts and comments from pre-approved Malawian public Facebook pages via the Apify FacebookCommentScraper API on a scheduled basis. All retrieved records shall be persisted in the database backend with their timestamps, source page, and identifiers.
FR2	Preprocess collected comment text prior to classification.	The system shall preprocess collected comment text to produce normalized, classification-ready input. Preprocessing shall remove noise, resolve language inconsistencies, and normalize dialectal and slang variations such that classification accuracy is not degraded by text artifacts.
FR3	Classify the sentiment of facebook comments.	The system shall automatically assign each comment one of three sentiment labels (<i>positive</i> , <i>neutral</i> , or <i>negative</i>) . Classification results shall be persisted in the database alongside the original comment record.
FR4	Display sentiment trends	The dashboard shall display time-series visualizations of sentiment distribution per topic, allowing users to observe how public opinion on a given topic shifts over days or weeks.
FR5	Filter dashboard views.	The dashboard shall allow users to filter visualizations by sentiment and date range.
FR6	Display example posts per sentiment	The dashboard shall surface a sample of posts or comments from each sentiment category to provide qualitative context for the quantitative trend data.
FR7	Handle processing errors gracefully	The system shall log failures in data retrieval, preprocessing, or classification without interrupting overall system operation. The dashboard shall display the last available results and a status notice if any component is temporarily unavailable.

FR8	Export analysis results	The system shall allow users to export filtered sentiment analysis results and dashboard data in CSV and PDF formats.
FR9	Display system status and data freshness	The system shall display system status information including the most recent data update timestamp to indicate dataset freshness and system availability.
FR10	Comment Drill-Down	The system shall allow users to click on dashboard visualizations, such as time-series points, to view the underlying Facebook comments associated with that data point.

2.3 System Interfaces

Table 6: System Interfaces

Interface Type	Description
Software Interface	Integration with PostgreSQL for structured storage
User Interface	Web-based dashboard using Streamlit or Dash
Data Interface	Manual export (CSV, JSON) as fallback
internal Software Interface	Use of Scikit-learn or Hugging Face Transformers

2.4 Interface Requirements

The table below defines all interfaces through which users, data, and external systems interact with the delivered system. Requirements are stated for the intended end system. The current prototype implementation is described separately in section 2.1.4.

Table 7: Interface Requirements

Req.	Interface Type	Description	External System
IR1	Data Collection Interface	The system shall retrieve public posts and comments from pre-approved Malawian Facebook pages via the Apify API on a scheduled basis. Retrieved records shall include text, timestamp, source page, and post/comment identifier. All ingested data shall be written to the database backend. Where API access is restricted, manually exported CSV files conforming to the required schema shall serve as the fallback ingestion method.	Apify API
IR2	Database Interface	The system shall persist all ingested posts, comments, preprocessing outputs, sentiment labels, topic labels, timestamps, and source metadata in a PostgreSQL or MongoDB database. The database shall serve as the central data store for all system components including data ingestion, preprocessing, classification, and dashboard presentation.	PostgreSQL / MongoDB
IR3	Classification Interface	The system shall apply pre-trained sentiment and topic classification models to preprocessed records retrieved from the database. Classification results shall be written back to the database and made available to the dashboard. The deployed models shall be loaded from a designated model storage location at system startup.	Internal (database-backed inference)
IR4	Model Storage Interface	The system shall load pre-trained sentiment and topic model artifacts from a designated storage location at startup. The system shall support swapping the deployed model to a newer version without requiring changes to other system components.	Local server / external model store

IR5	Visualization Dashboard Interface	The system shall provide an interactive web dashboard implemented using Streamlit or Dash. The dashboard shall present topic-wise sentiment trends over time, topic distribution charts, time-series sentiment analysis, and representative example posts per sentiment category. Users shall be able to filter results by topic and time period.	Web browser (Streamlit / Dash)
IR6	Data Collection Fallback Interface	Where Apify API access is unavailable, the system shall accept manually prepared CSV files as a fallback data ingestion method. Such files shall conform to the required schema including the fields: id, text, and include. A custom lexicon file shall be maintained to support Malawian slang normalisation during preprocessing.	File system (manual CSV export)

2.5 Non-Functional Requirements

System-Related Non-Functional Requirements

Table 8: System NFR

Req.	Interface Type	Description	External System
NFR1	Data anonymization	Extract text, reactions, and post ID only; exclude personal data	FR1
NFR2	Human alignment	Output should align with human evaluation	FR2
NFR3	Linguistic Robustness	Handle sarcasm, emojis, and code-mixing	FR2
NFR4	Ethical Compliance	Data used only for sentiment analysis; not shared raw	FR1
NFR5	Model Performance	Evaluate using Accuracy, Precision, Recall, F1-score	FR3
NFR6	Interpretability	Convert text into clear visual trends	FR5

User-Related Non-Functional Requirements

Table 9: User related NFR

Req.	Requirement	Description
UR1	Contextual understanding	Provide sample posts per sentiment category
UR2	Usability and User Testing	Dashboards validated through stakeholder testing

3. Data Requirements

3.1 Data Collection / Ingestion

The data for this project will consist of public Facebook comments related specifically to economic discussions in Malawi. Public Facebook comments will be collected from selected Malawian Facebook pages such as *Times 360 Malawi*, *Zodiac Online*, *Malawi24*, and *Mikozi* using a web scraping tool, Apify. Only publicly accessible posts and comments will be retrieved in compliance with ethical and privacy requirements.

The scraping process will collect:

- Comment text
- Comment ID
- Timestamp
- Source page information

The collected data will initially be stored in JSON format and later converted into CSV format for preprocessing and annotation.

To improve annotation efficiency and reduce unnecessary manual labeling effort, the project will adopt an active learning approach during dataset expansion. Instead of labeling all collected comments randomly, an initial manually labeled dataset will first be used to train a preliminary sentiment classification model. The model will then identify comments with the highest prediction uncertainty, and those comments will be prioritized for human annotation. This approach reduces annotation workload while improving model learning efficiency, especially in low-resource language settings such as Chichewa-English code-mixed text.

The annotation process will therefore follow these stages:

- Collection of raw Facebook comments
- Manual labeling of an initial seed dataset
- Initial model training

- Selection of uncertain comments through active learning
- Additional annotation and retraining

The dataset will contain approximately 2,000–5,000 manually reviewed comments during the development phase, with the possibility of future expansion through iterative annotation cycles.

3.2 Data Description

The dataset consists of user-generated Facebook comments related to economic topics.

Source: Facebook public posts (e.g., Times 360 Malawi, Mikozi, Zodiak Online; scraped using Apify/Scrapy)

Format: JSON (initially), later converted to CSV

Dataset variables include:

- **id (integer):** Unique identifier for each comment
- **text (string):** The actual facebook comment text
- **sentiment_label (string):** Positive, Negative, or Neutral
- **include (string):** Yes or No (indicator showing whether the comment should be included in the final dataset or not)
- **notes (string):** Optional annotator remarks explaining exclusions or ambiguities

3.2.1 Annotation Guidelines

To maintain annotation consistency and label quality, annotators will follow a structured annotation protocol;

Sentiment Labels

Each included comment will be assigned one sentiment label:

- **Positive:** Praise, approval, optimism, or satisfaction
- **Neutral:** Factual statements or comments without strong emotion
- **Negative:** Complaints, criticism, anger, or dissatisfaction

Exclusion Rules

A comment will be excluded (`include = No`) if:

- The sentiment is unclear or highly ambiguous
- The comment is difficult to interpret
- The text is irrelevant or off-topic
- The text contains excessive noise or incomplete information

The annotation principle adopted is:

“When in doubt, leave it out.”

Annotation Quality Control

To ensure reliable labels, the project will use:

- Multiple annotators for selected samples
- Annotator calibration exercises
- Inter-Annotator Agreement (IAA) evaluation using Cohen’s Kappa

The study will target:

- Cohen’s Kappa ≥ 0.70 as the acceptable annotation agreement threshold

If agreement falls below the threshold, annotation rules will be reviewed and clarified before continuing.

Annotation Tooling

The annotation process may be managed using spreadsheet-based labeling or annotation tools such as *Label Studio* to maintain structured annotation records.

Sample Size

The dataset is expected to contain approximately:

- 2,000–5,000 comments after preprocessing and filtering

Additional comments may later be incorporated through active learning and iterative annotation.

3.2.1 Limitations

- **Mixed language (Chichewa + English):** Many comments contain English-Chichewa code-mixing, which creates challenges for standard NLP tools designed for single-language processing.
- **Informal Text and Slang:** The dataset contains informal abbreviations, local slang, emoji’s, and internet expressions that may carry sentiment information but are difficult to normalize consistently.
- **Noisy data :** Some comments contain spam, unrelated discussions, repeated content, or unclear text that can reduce model quality if not removed.
- **Spelling variations and errors:** Chichewa words often appear with inconsistent spelling variations, abbreviations, and phonetic typing styles, making normalization difficult.
- **Class imbalance:** Some sentiment categories may naturally occur more frequently than

others, potentially causing model bias during training.

3.3 Exploratory Data Analysis

Exploratory Data Analysis (EDA) will be performed to understand the structure and characteristics of the dataset. This includes analyzing the distribution of sentiment labels, identifying frequently used words, and examining text length.

The analysis will also detect missing values, duplicates, and inconsistencies in the dataset. Visualization tools such as bar charts and word frequency plots will be used to identify patterns and trends.

Libraries such as pandas, matplotlib, and seaborn will be used to support the analysis.

3.4 Data Cleaning and Preprocessing

The system will perform data cleaning and preprocessing to ensure the dataset is consistent, accurate, and suitable for analysis and modeling.

The planned preprocessing steps include:

- **Handling missing values:** Comments with missing or empty text fields will be removed or handled appropriately.
- **Removal of duplicates:** Duplicate comments will be identified and removed to prevent training bias.
- **Noise handling:** Extra spaces, HTML tags irrelevant characters, and formatting inconsistencies will be cleaned from the text.
- **Text normalization:** All text will be converted to lowercase to maintain consistency during tokenization and vectorization.
- **Tokenization:** Text will be split into smaller units (tokens).
- **Stop word removal:** Common words (e.g., “the”, “is”) with limited semantic contribution may be removed where appropriate.
- **Stemming and Lemmatization:** Words may be reduced to their base or root forms to improve consistency across similar expressions.
- **Vectorization and Embedding:** Text will be converted into numerical format using Bag of Words or TF-IDF. Then transformer-based models such as AfriBERTa will use contextual embeddings generated directly from tokenized text.
- **Emoji handling (emoji to text conversion):** Emoji’s will be converted into textual representations because they often carry sentiment information.
- **Slang and Abbreviation Normalization:** A custom Malawian slang lexicon will be maintained to normalize common informal expressions and abbreviations. e.g “2bad” → too bad.

- **Code-Mixed Text Handling:** The preprocessing pipeline will support English-Chichewa mixed text by preserving meaningful multilingual expressions instead of aggressively removing non-English terms.
- **Data Augmentation:** To improve model generalization in this low-resource setting, the project may apply limited data augmentation techniques such as back translation, synonym replacement and minor text perturbations. These methods may help improve sentiment classification performance for underrepresented patterns.
- **Active Learning Integration:** The preprocessing pipeline will support iterative dataset expansion through active learning. Newly collected comments identified as highly uncertain by the model may be prioritized for annotation and later reintegrated into the training dataset.

3.5 Data Summary

The dataset used for this project consists of labeled facebook comments for sentiment classification within Malawi's economic discussion context.

- **Dataset size:** Approximately 2000-5000 comments
- **Type:** Unstructured multilingual text
- **Language:** English, Chichewa, and code-mixed variants
- **Classification task:** Sentiment Analysis

Key variables:

- Text (input feature)
- Sentiment label (target variable)
- Include flag
- Annotation notes

The dataset is suitable for training classification models to predict sentiment from text data.

4 Technical Requirements

4.1 Software and Tools

The system will be implemented using the Python data science ecosystem and modern NLP tooling suitable for multilingual sentiment analysis and dashboard visualization.

- Programming Language: Python 3.10+
- Development Environment: Jupyter Notebook, VS Code
- Data Handling Libraries: pandas, numpy, regex, emoji
- Natural Language Processing: nltk, transformers

- Machine Learning: scikit-learn (Logistic Regression, Naïve Bayes)
- Transformer Models: AfriBERTa via the Hugging Face Transformers library
- Database Connectivity: psycopg2 / pymongo
- Database Systems: PostgreSQL or MongoDB
- Dashboard Framework: Streamlit or Dash
- Visualization Libraries: matplotlib, plotly
- Version Control: Git and GitHub
- Data Collection Interface: Apify API for retrieving public posts and comments

These tools support data collection, preprocessing, model training, evaluation, storage, and visualization within a single integrated environment.

4.2 Hardware Requirements

The project does not require specialized hardware but benefits from moderate computational resources due to transformer model fine-tuning.

- Local Machine Requirements:
 - Minimum 8 GB RAM (16 GB recommended)
 - Intel i5 / AMD Ryzen 5 processor or equivalent
 - At least 20 GB free storage for datasets and models
- GPU Support (Optional):
 - NVIDIA GPU with CUDA support for faster transformer training
- Cloud Computing (Optional):
 - Google Colab, AWS, or university server for model training if local resources are limited
- Internet Connectivity:
 - Required for data retrieval, library installation, and dashboard access

These requirements ensure the system can efficiently process Facebook text data, train machine learning models, and serve interactive visual dashboards.

5. Ethics and Privacy

Following the Malawi Data-Protection-act 2024 [10], the system will adhere to ethical standards and data privacy principles throughout its development and deployment. Since the project involves the collection and analysis of user-generated content from Facebook, it is essential to ensure responsible and lawful handling of data.

- The system will only collect publicly available Facebook posts and comments from selected pages. No private messages, restricted groups, or personal accounts will be accessed. This ensures compliance with platform policies and respects user consent and data ownership.
- All collected data will undergo anonymization to protect user identity. Personally identifiable information such as usernames, profile links, and user IDs will be removed before analysis. The system will retain only the textual content necessary for sentiment and topic analysis.
- The project acknowledges the potential risk of bias in machine learning models. Since the dataset includes multilingual and code-mixed text (e.g., English and Chichewa), the model may misinterpret certain expressions or slang. To address this, the system will use diverse training data and apply careful evaluation techniques to improve fairness and accuracy.
- The system will ensure the ethical use of analytical results. The outputs generated, such as sentiment trends and topic distributions, will be presented as general insights rather than absolute truths. Users of the system will be informed that the results are predictive and may contain limitations.
- The project will comply with institutional and research ethics guidelines, ensuring that all data is securely stored and used strictly for academic purposes. These measures collectively ensure that the system operates in a responsible, transparent, and ethical manner.

6. Project Plan

This section outlines the strategy for developing, managing, and delivering the proposed sentiment analysis system.

6.1 Development Cost

The estimated development cost for the system is relatively low, as the project primarily relies on open-source tools and technologies. Costs may include internet access for data collection, computing resources, and optional hosting services for the deployment of the dashboard. The overall estimated cost is approximately MWK 100,000 to MWK 150,000, covering development, testing, and documentation activities.

6.2 Software Life Cycle Constraints

The project will adopt an Agile Software Development Life Cycle (SDLC) approach. This methodology allows for iterative development, continuous testing, and gradual improvement of system components.

The development process will be divided into phases, including:

- Data collection and preprocessing

- Model development and training
- System integration
- Testing and evaluation
- Final deployment

The Agile approach is suitable for this project because it provides flexibility in handling challenges such as multilingual data processing and model performance improvements.

6.3 System Delivery Specifications

The final system will be delivered as a web-based application developed using Python frameworks such as Flask or Streamlit.

The key deliverables will include:

- A trained sentiment and topic classification model.
- An interactive visualization dashboard.
- explore topic-based trends through visual representations.

6.4 Installation and Access Requirements

The system will be designed to operate on a local machine or a cloud-based server. Developers will manage the source code through a version control platform such as GitHub.

Users will access the system through a web browser interface, where they can interact with the dashboard and view analysis results. Initial deployment may involve a pilot phase with limited users before full implementation.

6.5 Project Timeline

The development of the system will follow a structured timeline to ensure timely completion of all tasks.

Table 10: Project Timeline

Phase	Dates	Activity
SEMESTER 1		
	27 -31 Oct 2025	Topic submission and Supervisor Allocation
	3 – 14 Nov 2025	Literature Review and Requirements Gathering
	17 – 21 Nov	Definition of Scope, Objectives and Requirements
	24 – 28 Nov	Proposal Presentations
	1 – 5 Dec 2025	Technological Tool and Framework Selection
	8 – 19 Dec 2025	Data collection (Apify)
	5 – 9 Jan 2026	Data Preprocessing and cleaning
	12 Jan 2026	Final Proposal Submission
SEMESTER 2		
	5 May 2026	Requirements Specification Submission
	5 – 23 May 2026	Data annotation and Feature Engineering
	2 – 20 June	Model Development (baseline and Fine-tuning)
	23 Jun – Jul 2026	Evaluation and Analysis (Integration Testing)
	7 – 11 Jul 2026	Dashboard Development and Final Presentation
	14 Jul 2026	Documentation and Final report

This timeline ensures a systematic progression from data acquisition to final system delivery.

7. Appendices

7.1 Sample Data Schema

The system will store collected and processed social media comments together with their predicted sentiment labels and metadata.

Sample Comment Dataset Schema

Table 11: Sample Comment Dataset Schema

Field Name	Data Type	Description
comment_id	Integer	Unique identifier for each comment
comment_text	Text	Original Facebook comment or post
cleaned_text	Text	Preprocessed version of the comment
language_type	Varchar(20)	Indicates whether text is Chichewa, English, or code-mixed
sentiment_label	Varchar(20)	Predicted sentiment category (Positive, Negative, Neutral)
date_collected	DateTime	Date and time the comment was collected

Example Record:

id	text	sentiment_label	include	notes
0	This is a major breakthrough for Malawi. Local production of fertilizer means food security, reduced import costs	Positive	Yes	
1	Someone is working hard to get a degree so that he can be employed by Dzombe in his company while Dzombe	Positive	Yes	
2	That guy he will be our malawian Elon musk	Positive	Yes	A comparison to a big public figure
3	Napoleon Grasshoppers, thinks like AI	Neutral	Yes	Doesnt make sense
4	If Napoleon,Simbi,Mpinganjira and Bushiri were to be well used,Malawi would not be the same again.	Neutral	No	
5	We want more Napoleon Dzombes than Politicians.	Positive	Yes	They are showing support by asking
6	When money meet vision	Positive	No	
7	Zeru zabambo	Positive	Yes	
8	While others are competing for leadership	Positive	Yes	
10	bwana mpinganjira atangomanga ena awiri we won't have to worry about fertiliser again and also unemployment	Positive	Yes	
11	Awawa man aaaaa I don't knw how to describe him ,so creative,so innovative,a true meaning of an entrepreneur	Positive	Yes	
12	Proud member of Dowia Nee Lumbadzi	Neutral	Yes	
13	The leader who had no tittle	Positive	Yes	
14	Katundu mzito Achalume,munthu mulala chomene	Positive	Yes	
15	Apapa wainyamula Malawi since Malawi's economy is Agriculture-based. We expect increased GDP per capita, s	Positive	Yes	
16	Uyuyu osazamutola	Positive	Yes	
17	I think our politicians can learn a bog lesson this man!	Positive	Yes	
	Kkk by the end of the day umva boma layamba kumukankza forex to order inputs for making fertilizer we have to			

Figure 1: Example Dataset Record

7.2 Evaluation Metrics

The performance of the sentiment analysis system will be evaluated using standard machine learning evaluation metrics including Accuracy, Precision, Recall, and F1-score. These metrics will be used to assess the effectiveness of the model in classifying sentiments from Chichewa-English code-mixed social media text.

To determine whether the system meets the required quality standards, the following minimum and target performance thresholds shall be used.

Table 13: Evaluation Metrics and Performance Thresholds

Metric	Description	Minimum Threshold	Target Threshold
Accuracy	Measures overall correctness of predictions	$\geq 70\%$	≥ 0.75
Precision	Measures correctness of positive predictions	≥ 0.65	≥ 0.75
Recall	Measures ability to identify relevant sentiments	≥ 0.65	≥ 0.75
F1-score	Harmonic mean of Precision and Recall	≥ 0.70	≥ 0.80

The system shall be considered acceptable for deployment and testing if the minimum thresholds are achieved. Target thresholds represent the desired performance goals for optimal system effectiveness.

Glossary

Table 11: Glossary of Key Terms

Term	Definition
Active Learning	A machine learning approach where the model selects uncertain samples for human annotation to improve learning efficiency.
Annotation	The process of manually assigning labels to text data for supervised machine learning.
Code-Mixing	The use of multiple languages within the same sentence or text, such as Chichewa-English mixed comments.
Cohen's Kappa	A statistical metric used to measure agreement between annotators beyond chance.
Feature Engineering	The process of creating or selecting useful input features from raw data for machine learning models.
Fine-Tuning	The process of adapting a pretrained machine learning model to a specific task or dataset.
Inference	The process where a trained machine learning model predicts outputs for new input data.
Inter-Annotator Agreement (IAA)	A measure of consistency between multiple annotators labeling the same dataset.
Metadata	Additional information describing data, such as timestamps, source page, or identifiers.
Pretrained Model	A machine learning model previously trained on large datasets before being adapted to another task.
Tokenization	The process of splitting text into smaller units such as words or subwords before model processing.
Transformer Model	A deep learning model architecture designed for advanced natural language processing tasks.
AfriBERTa	A transformer-based language model optimized for African languages to improve multilingual NLP performance.
Web Scraping	The automated extraction of publicly available data from websites or online platforms.

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